

Executive Summary: The AI Risk Repository

The Core Problem

The conversational and regulatory landscape surrounding AI risk is fragmented by inconsistent terminology (the "jingle-jangle fallacy"), where different terms are used for the same risk, or the same word defines completely separate problems. This confusion impedes coordinated regulatory, audit, and governance responses across international jurisdictions.

The Solution

The authors built the **AI Risk Repository**, a comprehensive, living database of **1,725 distinct risks** extracted from 74 existing taxonomies and frameworks. To make this actionable, they synthesized these risks into two complementary, intersecting frameworks: a **Causal Taxonomy** and a **Domain Taxonomy**.

1. The Causal Taxonomy (Origins of Risk)

The Causal Taxonomy addresses *how, when, and why* risks emerge by examining antecedent conditions. Every risk is classified under exactly one level for three core variables:

- **Entity:** Is the risk caused by human decisions (e.g., poor data selection, malicious deployment) or AI system actions (e.g., unexpected autonomous behaviors)?
- **Intent:** Is the risk an *intentional*, expected outcome of pursuing a goal (e.g., state surveillance), or an *unintentional*, unexpected consequence (e.g., a model hallucination due to biased training data)?
- **Timing:** Does the risk materialize *pre-deployment* (during training/coding) or *post-deployment* (when used by end-users)?

Key Statistical Intersections

- **Post-Deployment Dominance:** A notable 62% of all cataloged risks are focused on post-deployment consequences, while only 13% address development-phase vulnerabilities.
- **Human vs. AI Split:** Risks are nearly equally split between being caused directly by AI actions (42%) and human decisions (38%).
- **The Most Common Triad:** The most frequent risk profile found in the literature is **Human + Intentional + Post-deployment** (18% of all risks, e.g., fraud or cyberattacks), closely followed by **AI + Unintentional + Post-deployment** (14%, e.g., robustness failures).

2. The Domain Taxonomy (Types of Harm)

The Domain Taxonomy captures consequent hazards and impacts, organizing them into **7 overarching domains** and **24 subdomains**. Unlike the Causal Taxonomy, these domains are interconnected, and real-world incidents can span multiple categories.

Domain	Key Subdomains Covered	Prevalence in Literature	Legal/Regulatory Focus PDF+ 1
1. Discrimination & Toxicity	Unfair discrimination, exposure to toxic content, unequal group performance.	Present in 70% of documents	Direct relevance to civil rights, anti-bias frameworks, and human rights law.
2. Privacy & Security	Leaking sensitive data, inference without consent, toolchain/API vulnerabilities.	Present in 67% of documents	Critical for GDPR compliance, corporate data governance, and liability for data breaches.
3. Misinformation	Inadvertent false information generation, pollution of information ecosystems.	Present in 47% of documents	Impacts consumer protection law, free speech laws, and democratic protections.
4. Malicious Actors & Misuse	Scaled disinformation, automated cyberattacks, bioweapon assistance, consumer scams.	Present in 73% of documents	Falls under criminal law, national security regulation, and tort liability for software misuse.

5. Human-Computer Interaction	Overreliance/psychological dependence, loss of human agency/cognitive enfeeblement.	Present in 52% of documents	Important for defining user autonomy, product safety, and safe operation mandates.
6. Socioeconomic & Environmental	Power centralization, job devaluation, structural inequality, failure of regulatory pacing.	Present in 77% of documents	Directly intersects with anti-trust law, copyright/intellectual property lawsuits, and labor laws.
7. AI System Safety & Failures	Goal misalignment, dangerous capabilities, lack of explainability, agentic risks.	Present in 78% of documents	Essential for defining "strict liability" parameters and evaluating compliance audits.

3. Critical Gaps in Current AI Risk Frameworks

The meta-review exposed major deficiencies in how current academic and corporate frameworks conceptualize risk:

- **Narrow Frameworks:** The average existing framework only addresses 8 out of the 24 necessary subdomains. This leaving massive operational blind spots for compliance management.
- **The "Pre-Deployment" Blind Spot:** Because 62% of focus is placed on post-deployment harms, regulators and companies are systematically ignoring development-phase human risks (such as insecure coding structures and uncalibrated reward-modeling).
- **Neglected Emerging Threats:** Novel areas such as **Multi-agent risks** (cascading failures when AI systems interact autonomously with other AIs) are featured in only 7% of current frameworks. **AI welfare and rights** (sentience considerations) appear in a tiny 3% of literature.

4. Legal & Regulatory Implications

The repository provides immediate utility in addressing three fundamental legal dilemmas:

The Responsibility & Accountability Gap

Traditional legal structures require identifying a human actor's intent or negligent action to assign fault. Because AI systems utilize highly opaque, non-linear deep learning pathways, pinpointing proximate cause is exceptionally difficult. Regulators can use the repository's **Causal Taxonomy** to distinguish where liability should lie—for example, targeting the 37% of human-attributed risks via professional training mandates, while enforcing strict technical standard compliance for the 42% of intrinsic AI system risks.

Operationalizing Vague Statutes

Statutes like the **EU AI Act** heavily rely on highly abstract phrases like "high-risk systems" or "harm." The AI Risk Repository solves this translation problem by offering **1,725 explicit, empirically recorded risk examples** that can step-by-step convert vague statutory concepts into concrete compliance checklists and standard operating audit procedures.

Intellectual Property & Regulatory Lag

Subdomain 6.3 highlights the legal "gray area" regarding generative AI utilizing copyright-protected materials without explicit authorization to systematically devalue human effort. The paper emphasizes that regulatory processes are inherently too slow to keep pace with these shifts. The living, extensible nature of this database allows legal scholars and compliance officers to continuously track newly emerged categories of civil and economic injury before standard statutory revision cycles capture them.